Global seroprevalence and sociodemographic characteristics of *Borrelia burgdorferi sensu lato* in human populations: A systematic review and meta-analysis

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Supplementary appendix 1. PRISMA 2020 checklist

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	√, title
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	√, abstract
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	√, line 91-116
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	√, line 118-128
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	√, line 149-181
Information sources	6	Specify all databases, registers, websites, organizations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	√, line 118-128
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	√, line 135-147
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	√, line 168-172
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in	√, line 175-177

Section and Topic	Item #	Checklist item	Location where item is reported
		the process.	
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	√, line 177-181
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	√, line 145-147
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	√, line 183-199
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	√, line 201-223
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	√, line 201-228
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	√, line 212-223
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	√, line 207-222
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	√, line 202-210
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	√, line 212-223
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	√, line 220-222
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	√, line 183-199
Certainty	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	√, line 222-224

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	√, fig 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	√,Supplementary Appendix 3
Study characteristics	17	Cite each included study and present its characteristics.	√, Supplementary Appendix 5
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	√, Supplementary Appendix 6
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	√, line 118-128
Results of syntheses	20a	For each synthesis, briefly summarize the characteristics and risk of bias among contributing studies.	√, table 1,2,3
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	√, line 242-300
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	√, line 242-312
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	√, line 302-307
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	√, line 314-317

Section and Topic	Item #	Checklist item	Location where item is reported
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	√, line 304-307
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	√, line 327-406
	23b	Discuss any limitations of the evidence included in the review.	√, line 334-357
	23c	Discuss any limitations of the review processes used.	√, line 408-425
	23d	Discuss implications of the results for practice, policy, and future research.	√, line 430-444
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	√, line 463-464
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	√, line 463-467
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	√, line 464-467
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	√, line 459-463
Competing interests	26	Declare any competing interests of review authors.	√, line 468
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	√, line 471-489

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

For more information, visit: http://www.prisma-statement.org/

Supplementary appendix 2. Search strategy

PubMed (https://www.ncbi.nlm.nih.gov/pubmed/) (1984 to 21 Nov 2021)

#1	*Borrelia burgdorferi* [Mesh] OR Lyme Disease Spirochete [Title/Abstract] OR *Borrelia burgdorferi sensu stricto* [Title/Abstract]
#2	infection rate [Title/Abstract] OR prevalence [Title/Abstract] OR seroprevalence [Title/Abstract] OR serological surveys [Title/Abstract] OR sero-prevalence [Title/Abstract] OR seroepidemiology [Title/Abstract] OR sero-epidemiology [Title/Abstract]
#3	#1 AND #2

EMBASE (https://www.embase.com/login) (1984 to 18 Dec 2021)

#1	'borrelia burgdorferi'/exp
#2	'borrelia burgdorferii' OR 'borreliella burgdorferi' OR 'lyme disease spirochete' OR 'borrelia burgdorferi sensu stricto' OR 'borrelia burgdorferi test kit'
#3	#1 OR #2
#4	'seroepidemiology'/exp
#5	'sero-epidemiology' OR 'seroepidemiologic methods' OR 'seroepidemiologic studies' OR 'serogroup epidemiology' OR 'serologic epidemiology' OR 'serological epidemiology' OR 'serotype epidemiology' OR 'serotypic epidemiology' OR 'sero-prevalence' OR 'serological surveys' OR 'seroprevalence' OR 'prevalence' OR 'infection rate'
#6	#4 OR #5
#7	#3 AND #6

Web of Science (https://www.webofscience.com/wos) (1991 to 30 Dec 2021)

#1	*TS= (Borrelia burgdorferi sensu stricto)*
#2	TS= (Lyme Disease Spirochete)
#3	#1 OR #2

Sources of Grey Literature Search

Search strategy: Reports, surveys, censuses and vital statistics using the search terms "*Borrelia burgdorferi sensu stricto*" OR "Lyme Disease" OR "anti-*Bb*" OR "*Bb*" OR "infection rate", and screened all articles with the keyword "*Borrelia burgdorferi*" within the following sources:

1. Global Health Data Exchange (http://ghdx.healthdata.org/)

2. The Demographic and Health Survey Programmes (https://www.dhsprogram.com/)

3. World Bank Living Standards Measurement Survey (http://iresearch.worldbank.org/lsms/lsmssurveyFinder.htm)

Supplementary appendix 3. Exclude literatures and reasons (N=48)

Reason 1: Full-text unavailable, or no necessary data (N=18)

1. G. I. Kritchevskaya, E. S. Vakhova, O. B. Vylivannaya, G. A. Davydova. The rate of detection and diagnostic significance of antibodies to *Borrelia burgdorferri* in patients with eyes diseases of inflammatory nature. Klin Lab Diagn 2014(2):53-56. pmid:2506923520014-02-01.

2. Heroldová M. Dvořáková, M. Dvořáčková. Seroprevalence of Anaplasma phagocytophilum in patients with suspected Lyme borreliosis. Epidemiol Mikrobiol Imunol 2014;63(4):297-302. pmid:255232232014-11-01.

3. A. Vatne, A. Mygland, U. Ljøstad. Multiple sclerosis in Vest-Agder County, Norway. ACTA NEUROL SCAND 2011;123(6):396-99. doi: 10.1111/j.1600-0404. 2010.01411.x pmid: 21492097. © 2010 John Wiley & Sons A/S.*2011-06-01.

4. G. Biesiada, J. Czepiel, D. Salamon, et al. Analysis of *Borrelia burgdorferi* genostrains among patients with Lyme disease. Przegl Lek 2009;66(9):511-12. pmid:210334122009-01-20.

5. Sevgi Ergin. Molecular detection and prevalence of *borrelia sensu lato* genotypes in human biting ticks in Istanbul, Turkey. International Union of Biochemistry & Molecular Biology Life 2009;61(3):318.

6. M. Livio, A. Mobilia, S. Abbate, et al. Ticks bite in foresters. G Ital Med Lav Ergon 2007;29(3 Suppl):811-12. pmid:184099752007-07-01.

7. D. Dybowska, D. Kozielewicz, A. Abdulgater, et al. Prevalence of borreliosis among forestry workers in Kujawsko-Pomorskie voivodeship. Przegl Epidemiol 2007;61(1):67-71. pmid:177024412007-01-20.

8. J. Walory, B. Bukowska, P. Grzesiowski, et al. Prevalence of antibodies against Anaplasma phagocytophilum, Babesia microtii *Borrelia burgdorferi* in adults in North-Eastern Poland. Pol Merkur Lekarski 2005;19(114):754-57. pmid:165214162005-12-01.

9. P. Castiglia, I. Mura, M. D. Masia, et al. Prevalence of antibodies to *Borrelia burgdorferi* in Sardinian teen-agers. Ann Ig 2004;16(1-2):103-08. pmid:155545162004-01-01.

10. T. Chmielewski, S. Tylewska-Wierzbanowska. Prevalence of *Borrelia burgdorferi* antibodies in healthy population in Poland. Przegl Epidemiol 2002;56(1):33-38. pmid:121500652002-01-20.

11. V. Mrázek, P. Bartůněk, P. Varejka, et al. Prevalence of antiborrelia antibodies in two populations: various interpretations of the same data. Epidemiologie, mikrobiologie, imunologie: casopis Spolecnosti pro epidemiologii a mikrobiologii Ceske lekarske spolecnosti JE Purkyne 2002;51(1):19-22.

12. M. Cizman, T. Avsic-Zupanc, M. Petrovec, E. Ruzic-Sabljic, M. Pokorn. Seroprevalence of ehrlichiosis, Lyme borreliosis and tick-borne encephalitis infections in children and young adults in Slovenia. WIEN KLIN WOCHENSCHR 2000;112(19):842-45. pmid:110985352000-10-13.

13. R. Flisiak, D. Prokopowicz. Antibodies against Borrelia garinii in diagnosis of Lyme disease. Przegl Lek 2000;57(3):147-49. pmid:109092832000-01-20.

14. M. Cizman, T. Avsic-Zupanc, M. Petrovec, E. Ruzic-Sabljic, M. Pokorn. Seroprevalence of ehrlichiosis, Lyme borreliosis and tick-borne encephalitis infections in children and young adults in Slovenia. WIEN KLIN WOCHENSCHR 2000;112(19):842.

15. D. Golubić, S. Rijpkema, N. Tkalecmakovec, E. Ruzić. Epidemiologic, ecologic and clinical characteristics of Lyme borrelliosis in northwest Croatia. Acta Med Croatica 1998;52(1):7-13.

16. T. Weiland, P. Kühnl, R. Laufs, J. Heesemann. Prevalence of Borrelia burgdorferi antibodies in Hamburg blood donors. Beitr Infusionsther 1992;30:92-95. pmid:12847771992-01-19.

17. Bruinsma RA, Smulders CA, Vermeeren YM, van Kooten B, Cats EA, van Hees B, van Hensbroek MB, Hovius JW, Zomer TP. Acute facial nerve palsy in children in a Lyme disease-endemic area in the Netherlands. Eur J Clin Microbiol Infect Dis. 2021 Nov;40(11):2455-2458. pmid: 33977412.

18. Arnaboldi PM, D'Arco C, Hefter Y, Nolan S, Jobe DA, Callister SM, Dattwyler RJ. Detection of IFN-γ Secretion in Blood Samples Collected Before and After Treatment of Varying Stages of Lyme Disease. Clin Infect Dis. 2021 Oct 20;73(8):1484-1491. pmid: 34043758.

Reason 2: Unrelated data or article (N=16)

17. Artem S. Rogovskyy, Anton P. Biatov, Margaret Alison Davis, Shuling Liu, Igor V. Nebogatkin. Upsurge of Lyme borreliosis in Ukraine: a 20-year survey. J TRAVEL MED 2020;27(6):a100.

18. Tran Duc Anh Ly, Meriem Louni, Van Thuan Hoang, Thi Loi Dao, Philippe Gautret. Epidemiological serosurvey of vector-borne and zoonotic pathogens among homeless people living in shelters in Marseille: cross-sectional one-day surveys (2005–2015). European Journal of Clinical Microbiology 2020.

19. H. Pöyhönen, T. Lähdesmäki, J. Hytönen, V. Peltola. Cerebrospinal Fluid Pleocytosis and Elevated C-X-C Motif Chemokine Ligand 13 Value Predict Lyme Borreliosis in Children With Facial Palsy. PEDIATR INFECT DIS J 2019;38(12):1195-98. doi: 10.1097/INF.0000000000002475 pmid:316342972019-12-01.

20. J. Svensson, K. P. Hunfeld, KEM Persson. High seroprevalence of Babesia antibodies among Borrelia burgdorferi-infected humans in Sweden. Ticks Tick Borne Dis 2019;10(1):186-90. doi: 10.1016/j.ttbdis.2018.10.007 pmid:30389326. Copyright © 2018 The Authors. Published by Elsevier GmbH. All rights reserved. *2019-01-01.

21. E. Gałęziowska, J. Rzymowska, N. Najda, et al. Prevalence of *Borrelia burgdorferi* in ticks removed from skin of people and circumstances of being bitten - research from the area of Poland, 2012-2014. Ann Agric Environ Med 2018;25(1):31-35. doi: 10.5604/12321966.1233906 pmid:295758812018-03-14.

22. A. Alaedini, B. Lebwohl, G. P. Wormser, P. H. Green, J. F. Ludvigsson. Borrelia infection and risk of celiac disease. BMC MED 2017;15(1):169. doi: 10.1186/s12916-017-0926-1 pmid:289113262017-09-15.

23. P. H. Boyer, Sjd Martino, Y. Hansmann. No evidence of Borrelia mayonii in an endemic area for Lyme borreliosis in France. PARASITE VECTOR 2017;10(1).

24. A. C. Garro, M. S. Rutman, K. Simonsen, et al. Prevalence of Lyme meningitis in children with aseptic meningitis in a Lyme disease-endemic region. PEDIATR INFECT DIS J 2011;30(11):990-92. doi: 10.1097/INF.0b013e31822e8e68 pmid:219090502011-11-01.

25. S. Mavin, R. Evans, R. M. Milner, J. M. Chatterton, D. O. Ho-Yen. Local *Borrelia burgdorferi sensu stricto* and Borrelia afzelii strains in a single mixed antigen improves western blot sensitivity. J CLIN PATHOL 2009;62(6):552-54. doi: 10.1136/jcp.2008.063461 pmid:192400472009-06-01.

26. I. Tjernberg, G. Krüger, I. Eliasson. C6 peptide ELISA test in the serodiagnosis of Lyme borreliosis in Sweden. Eur J Clin Microbiol Infect Dis 2007;26(1):37-42. doi: 10.1007/s10096-006-0239-3 pmid:171803482007-01-01.

27. B. Mantelli, E. Pecchioli, H. C. Hauffe, R. Rosà, A. Rizzoli. Prevalence of *Borrelia burgdorferi s.l.* and Anaplasma phagocytophilum in the wood tick Ixodes ricinus in the Province of Trento, Italy. Eur J Clin Microbiol Infect Dis 2006;25(11):737-39.

28. T. Quessada, F. Martial-Convert, S. Arnaud, et al. Prevalence of *Borrelia burgdorferi* Species and Identification of Borrelia valaisiana in Questing Ixodes ricinus in the Lyon Region of France as Determined by Polymerase Chain Reaction-Restriction Fragment Length Polymorphism. EUR J CLIN MICROBIOL 2003;22(3):165-73.

29. A. Hildebrandt, K. H. Schmidt, B. Wilske, et al. Prevalence of four species of *Borrelia burgdorferi sensu lato* and coinfection with Anaplasma phagocytophila in Ixodes ricinus ticks in central Germany. EUR J CLIN MICROBIOL 2003;22(6):364-67.

30. A. Griskevicius, V. Lipnickiene, S. Caplinskas. Prevalence of *B. burgdorferi* strains by applying the microgen immunoblots. INT J MED MICROBIOL 2002;291 Suppl 33:224. doi: 10.1016/s1438-4221(02)80062-5 pmid:121417532002-06-01.

31. M. E. Aguero-Rosenfeld, L. Donnarumma, L. Zentmaier, et al. Seroprevalence of antibodies that react with Anaplasma phagocytophila, the agent of human granulocytic ehrlichiosis, in different populations in Westchester County, New York. J CLIN MICROBIOL 2002;40(7):2612-15. doi: 10.1128/JCM.40.7.2612-2615.2002 pmid:120892872002-07-01.

32. M. J. Mathiesen, K. Hansen, N. Axelsen, L. Halkier-S Rensen, M. Theisen. Analysis of the human antibody response to outer surface protein C (OspC) of *Borrelia burgdorferi sensu stricto,* B. garinii, and B. afzelii. Medical Microbiology & Immunology 1996;185(3):121-29.

Reason 3: Reviews, Meta, or lack eligible data in studies (N=4)

33. Zuzana Strizova, Daniel Smrz, Jirina Bartunkova. Seroprevalence of Borrelia IgM and IgG Antibodies in Healthy Individuals: A Caution Against Serology Misinterpretations and Unnecessary Antibiotic Treatments. Vector borne and zoonotic diseases (Larchmont, N.Y.) 2020;20(10).

34. K. Borawski, J. Dunaj, P. Czupryna, et al. Prevalence of Spotted Fever Group Rickettsia in North-Eastern Poland. Infect Dis (Lond) 2019;51(11-12):810-14. doi: 10.1080/23744235.2019.1660800 pmid:314963382019-11-01.

35. R. Bakker, M. C. Aarts, G. J. van der Heijden, M. M. Rovers. No evidence for the diagnostic value of Borrelia serology in patients with sudden hearing loss. Otolaryngol Head Neck Surg 2012;146(4):539-43. doi: 10.1177/0194599811432535 pmid:223945512012-04-01.

36. J. Simeoni, P. Conci, F. M. Broger, C. Nobile, G. Stinco. Epidemiology of Lyme borreliosis in Alpe Adria area. Acta Derm Venerol (Ljubljana) 1994;3:23-26.

Reason 4: Serum sample less than 30 or unused serum data (N=5)

37. G. Nijman, B. V. Kooten, Y. M. Vermeeren, Bcv Hees, T. P. Zomer. Antibiotic treatment in patients that present with solely non-specific symptoms and positive serology at a Lyme centre. EUR J INTERN MED 2020.

38. John SP Tulloch, Robert M. Christley, Alan D. Radford, et al. A descriptive epidemiological study of the incidence of newly diagnosed Lyme disease cases in a UK primary care cohort, 1998–2016. BMC INFECT DIS 2020;20(1):1-13.

39. M. Lager, R. B. Dessau, P. Wilhelmsson, et al. Serological diagnostics of Lyme borreliosis: comparison of assays in twelve clinical laboratories in Northern Europe. EUR J CLIN MICROBIOL 2019;38(10):1933-45.

40. Mercedes Treviño Castellano, Daniel Navarro de la Cruz, Rocio Trastoy Pena. *Borrelia burgdorferi* serology and diagnosis of Lyme disease in the healthcare area of Santiago de Compostela (Galicia, Spain): 2006-2016. MED CLIN-BARCELONA 2018;151(4):162-63.

41. G. Eldøen, I. S. Vik, E. Vik, R. Midgard. Lyme neuroborreliosis in More and Romsdal. Tidsskr Nor Laegeforen 2001;121(17):2008-11. pmid:118758962001-06-30.

Reason 5: Additional duplicates (N=5)

42. L. Lledó, M. I. Gegúndez, C. Giménez-Pardo, et al. A seventeen-year epidemiological surveillance study of *Borrelia burgdorferi* infections in two provinces of northern Spain. Int J Environ Res Public Health 2014;11(2):1661-72. doi: 10.3390/ijerph110201661 pmid:244874552014-01-30.

43. Islay Rodríguez, Carmen Fernández, Lizet Sánchez, Bárbara Martínez, Reto Lienhard. Prevalence of antibodies to *Borrelia burgdorferi sensu stricto* in humans from a Cuban village. Brazilian Journal of Infectious Diseases An Official Publication of the Brazilian Society of Infectious Diseases 2012;16(1).

44. E. Cisak, J. Chmielewska-Badora, J. Zwoliński, A. Wójcik-Fatla, J. Dutkiewicz. Risk of tick-borne bacterial diseases among workers of Roztocze National Park (south-eastern Poland). Ann Agric Environ Med 2005;12(1):127-32.

45. Ikushima, Yamada, Kawahashi, Okuyama, Matsui. Antibody response to OspC-I synthetic peptide derived from outer surface protein C of *Borrelia burgdorferi* in sera from Japanese forestry workers. EPIDEMIOL INFECT 1999;1999,122(3) (-):429-33.

46. Lourdes, Lledó, Consuelo, et al. Screening of Forestry Workers in Guadalajara Province (Spain) for Antibodies to Lymphocytic Choriomeningitis Virus, Hantavirus, Rickettsia spp. and *Borrelia burgdorferi.* INT J ENV RES PUB HE;16(22):4500.

Supplementary appendix 4. Quality assessment checklist：Newcastle-Ottawa scale adapted to cross-sectional studies.

Selection: (Maximum 5 stars)

1. Representativeness of the sample. (Representativeness of the sample)

a. Truly representative of the average in the target population. (multi-centre AND random sampling) **

b. Somewhat representative of the average in the target group. (non-random sampling AND/OR single centre) *

c. Selected group of users/convenience sample. (e.g., healthcare professionals, volunteers, etc.) may lead to significant selective bias.

d. No description of the sampling strategy.

2. Sample size (Sample size)

a. Justified and satisfactory. *

b. Less than 30 participants assessed for outcome.

c. No information provided.

3. Ascertainment exposure (risk factors) and assessment methods. (Ascertainment of exposure)

Exposure factors include both population type and general characteristics.

3.1 General characteristics: age, sex, geographic region, country, year of publication, season, tick bite status, detection method, etc.

3.2 Population classification.

A. General population (e.g., community samples, blood donors, physical examination population, general hospitalized population with non-Lyme disease symptoms).

B. Specific high-risk populations (associated with occupational exposure, e.g., farmers, herders, pastoralists, forest rangers, gardeners, border guards, military personnel, veterinarians, wildlife biologists, mountain/forest aborigines, hunters, game wardens, etc.).

C. Tick bitten people.

D. People with suspected Lyme disease-like symptoms (primary symptoms e.g., wandering erythema of the skin, flu-like symptoms, arthralgia, muscle pain, chest pain, neurological damage, cardiac abnormalities, chronic atrophic dermatitis, chronic arthritis, etc., secondary symptoms: sudden deafness, psychiatric symptoms).

a. Low risk of misidentifying exposure factors: rigorous, reliable medical records/structured questionnaires, thereby determining that the recruited population clearly falls into one of these categories. *

b. High risk of misidentifying exposure factors: Significant bias in recruiting population for retrospective data sessions, or low representation of the recruited population or significant inclusion exclusions and criteria that reduce applicability to the target population.

c. No information was provided.

Comparability: (Maximum 2 stars)

1. Comparability of subjects in different outcome groups on the basis of design or analysis. Confounding factors controlled. (Confounders)

a. Strictly control for bias due to important factors, such as the diagnostic method of serum *Bb* antibody used (kit/assay standard/assay facility/timing of sample testing, etc). *

b. The study controls for any additional factors, such as age, gender, population source area, population recruitment method. *

c. Data/results not adjusted for relevant confounders/risk factors/information not provided.

Outcome: (Maximum 3 stars)

1. Assessment of outcome. (Ascertainment of outcome)
a. Use standard serum Bb antibody detection techniques and clearly describe (including conventional serological methods such as Enzyme-Linked Immunosorbent Assay (ELISA), indirect fluorescent antibody test (IFA), protein biochip, chemiluminescence immunoassay, passive hemagglutination assays, line blot, and WB; laboratory molecular detection methods used as additional detection methods in some studies: polymerase chain reaction (PCR), real-time PCR, PCR–restriction fragment length polymorphism, and sequencing assays)， and the assay is consistent with the description, and is measured and evaluated independently and blindly, or repeatedly measured and evaluated for equivocal/negative results, or take some positive results for additional antibody assays (including tested for rheumatoid factor, Venereal disease research laboratory test (VDRL), and antinuclear antibodies) to exclude cross-reactivity etc. *
b. Use standard methods for detecting serum Bb antibodies that are clearly described and the results are reliably documented. *
c. No description/non-standard methods used/ The detection method is inconsistent with the description.
2. Non respondents
a. Proportion of target sample recruited attains pre-specified target (>80%) or basic summary of non-respondent characteristics in sampling frame recorded. *
b. Unsatisfactory recruitment rate, no summary data on non-respondents.
c. No information provided.
3. was there appropriate statistical analysis? (Statistical analysis)
a. Appropriate statistical methods were available and statistical analysis of the data was performed.
b. Statistical analysis was performed, but the method was not specifically described.
c. No statistical analysis was performed.

This scale was adapted from the Newcastle-Ottawa Quality Assessment Scale used in cohort studies and is appropriate for the themes of our systematic review and for conducting quality assessment in cross-sectional studies.
Quality classification criteria: the overall quality of the study was classified as low quality when the study scored less than 50%, moderate quality when the study reached a score of 50% to 69%, and high quality when the study reached a score of 70% or more.

Supplementary appendix 5. List of seroprevalence and potential risk factor in included 89 studies.

Country	Region	Method	Age(y)	Group	Positive/Total (%)	Determined risk factors	Author year
North America							
America	Maine	ELISA, conf. WB	8-90	Persons bitten by ticks	26/190 (13.68%)	N/A	Smith et al 2019[1]
				Family of database persons	6/40(15.00%)		
	California	ELISA, conf. WB	N/A	All population	24/60(40.00%)	N/A	Best et al 2019[2]
	California	ELISA	N/A	Blood donor	8/249(3.21%)	Tick bite History, age (older), time spent bird or wildlife watching or hiking (more)	Pascoe et al 2019[3]
	California	ELISA, conf. WB	16-80	Blood donor	8/1700(0.47%)	N/A	Brummitt et al 2020[4]
	Boston	ELISA, conf. WB	1–18	Patients with suspected LB	156/1066(14.63%)	N/A	Lipsett et al 2019[5]
	New York	ELISA, conf. WB	Adults	Veterans	181/409(44.25%)	Tick bite history, headache, joint ache/swelling,	Psevdos et al 2018[6]
	Massachusetts, Rhode Island, Pennsylvania, Wisconsin,	ELISA	1–21	Potential *Bb* infection population	556/1584(35.10%)	N/A	Maulden et al 2020[7]
Mexico	Mexico City, northeast of Mexico	ELISA, conf. WB	1-90	All population	122/2346(5.20%)	N/A	Gordillo et al 2003[8]
	Mexico City	ELISA, conf. WB	Adults	Patients with facial palsy	23/191(12.04%)	N/A	Gordillo-Pérez et al 2017[9]
Canada	Nova Scotia	ELISA, conf. WB	10-64	Partially	2/1855(0.11%)	N/A	Hatchette TF 2015[10]
Caribbean							
Cuba	Sierra del Rosario	WB	Adult	Inhabitants of a village	5/247(2.02%)	N/A	Rodríguez et al 2012[11]
South America							
Columbia	Cordoba	ELISA, conf. WB	16-70	Agricultural workers	7/152(4.61%)	N/A	Miranda et al 2009[12]
Brazil	Sao paulo	ELISA, conf. WB	N/A	Patients with sudden deafness	10/47(21.28%)	N/A	Lorenzi et al 2003[13]
	Sao paulo	ELISA, conf. WB	≤15	Children with skin rash and fever	12/193(6.21%)	< 6 years, female, city of Franco da Rocha, summer	Passos et al 2009[14]
Southern Europe							
Italy	North-east	ELISA, conf. WB	N/A	Forestry rangers	42/181(23.20%)	Employment in the foothills, history of tick bites, gardening in a northeast community	Cinco et al 2004[15]
	Tuscany	ELISA, conf. WB	N/A	Agricultural and forestry workers	64/412(15.53%)	Agricultural and forestry workers	Tomao et al 2005[16]
				healthy blood donors	25/365(6.85%)		

Country	Region	Method	Age(y)	Group	Positive/Total (%)	Determined risk factors	Author year
	Lazio	ELISA, conf. WB	N/A	Forestry workers	10/145(6.89%)	Outdoor tasks performed in rural environment, lack of individual protective devices, sex (male)	Di et al 2010[17]
				Blood donors	22/282(7.80%)	N/A	
	Etsch area	ELISA, conf. WB	18-61	Blood donors	10/597(1.50%)	N/A	Sonnleitner et al 2015[18]
Serbia	Belgrade	ELISA	N/A	Fieldworks	8/34(23.53%)	Fieldworks	sKrstić et al 2007[19]
				Control group	1/35(2.86%)		
	Belgrade	ELISA, conf. WB	25-45	Forestry workers	S4/30(13.33%)	Occupational exposure to *Bb* (soldiers and forestry workers)	Jovanovic et al 2015[20]
				Soldiers	6/35(17.14%)		
				Blood donors	3/35(8.57%)		
Slovenia	Ljubljana	CLIA	15-87	Patients diagnosed with PFP	120/436(27.52%)	Age(older), summer, previous LB, tick bites, constitutional symptoms (Malaise, Headache, Myalgia, Arthralgia, Radicular pain), bilateral palsy, CSF	Rojko et al 2019[21]
Western Europe							
United Kingdom	Bolton	ELISA, conf. WB	N/A	All population	100/250(40.00%)	N/A	Best et al 2019[2]
	Scotland	ELISA, conf. WB	N/A	Blood donors	60/1440(4.17%)	Residing in the Inverness postcode area	Munro et al 2015[22]
	Scotland	ELISA, conf. WB	N/A	All population	85/233(36.48%)	N/A	Mavin et al 2011[23]
France	South-western	ELISA, conf. WB	≥65	Retired farmers	45/689(6.53%)	Age (older), sex (men), CES-D scores (lower), LTDs (higher)	Ruiz et al 2020[24]
	Lyon	ELISA, conf. WB	6-92	Patients with uveitis	34/430(7.91%)	Sex (male)	Bernard et al 2017[25]
	Alsace, Lorraine, Champagne-Ardenne, Bourgogne, Franche-Comté	ELISA	15-81	Forest workers	419/2975(14.08%)	Living area (Alsace, Lorraine), age (≥45years), occupational categories (woodcutters)	Thorin et al 2008[26]
Sweden	Southeast area	ELISA	5	Healthy children	64/2000(3.20%)	Previous LD	Skogman et al 2010[27]
	Falun	ELISA/IFA	13-77	Blood donors	3/100(3.00%)	N/A	Madsen et al 2017[28]
				Patients with rheumatoid arthritis	5/60(8.33%)		
				Patients seeking general medical care	4/56(7.14%)		
Germany	Southwestern	ELISA	18-29	Male soldiers	19/361(5.26%)	N/A	Woessner et al 2001[29]
	Marburg	ELISA, conf. WB	N/A	Patients with suspected inflammatory heart	64/77(83.12%)	N/A	Karatolios et al 2014[30]
	Augsburg	ELISA, conf. WB	N/A	All population	113/177(63.84%)	N/A	Best et al 2019[2]

Country	Region	Method	Age(y)	Group	Positive/Total (%)	Determined risk factors	Author year
Netherlands	Berlin	ELISA, conf. WB	1-17	Children and adolescents	515/12614(4.08%)	Sex (male), age (older), geographical area, (southern part), migration background, residential area (rural areas or small towns), pet in household (cat)	Dehnert et al 2012[31]
	Berlin	ELISA, conf. WB	Adults	All population	654/6945(9.42%)	Sex (men), age (>59 years), residence in a rural area and in southern Germany, foreign national (no), having a	Wilking et al 2015[32]
	Nationwide	ELISA, conf. WB		General population	1225/12467(9.83%	Sex (male), age (70-79 years)	Woudenberg et al 2020[33]
	Freiburg	WB	Adults	Patients with Lyme neuroborreliosis	10/43(23.26%)	N/A	Dersch et al 2018[34]
				Healthy control	6/40(15.00%)		
	Utrecht	ELISA, conf. WB	20-91	Patients with ALS	20/491(4.07%)	N/A	Visser et al 2017[35]
				Controls	58/982(5.91%)		
	Utrecht	ELISA, conf. WB	N/A	Patients with Uveitis	56/1126(4.97%)	N/A	Kazi et al 2016[36]
	Apeldoorn	ELISA, conf. WB	18-87	Patients visited Lyme center	455/1439(31.62%)	N/A	Zomer et al 2018[37]
	Apeldoorn	ELISA, conf. WB	Adult	Patients with suspected LB	486/1454(33.43%)	N/A	Zwerink et al 2018[38]
Belgium	Walloon region, Namur, Brussels	ELISA/IFA	18-72	Professionally exposed workers	8/148(5.41%)	Sex (male), age (50+ years)	De et al 2017[39]
				Rural blood donors	6/209(2.87%)	Sex (female)	
				Urban blood donors	5/193(2.59%)	N/A	
	Walloon region	ELISA	24-65	Forest worker	67/310(21.61%)	Age (50+ years), duration of employment in silvicultural field (20+ years), forest visits (three times per week or more), number of tick bites (100+ times), frequency of tick bites (1 per month or more)	De et al 2018[40]
	Brussels, Flanders, Wallonia	ELISA, conf. WB	1-99	General population	37/3215(1.15%)	Sex (male), age (younger than 15, older than 59 years)	Lernout et al 2019[41]
Ireland	Western	ELISA, conf. WB	N/A	General population	242/5049(4.79%)	Summer, age(older)	Vellinga et al 2018[42]
Spain	Palencia, Burgos	IFA, conf. WB	3 mon	Patients bitten by ticks	83/623(13.32%)	N/A	Lledó et al 2014[43]
	Barcelona	ELISA	0.5-90	Patients in hospital	9/203(4.43%)	N/A	Segura et al 2004[44]
	Madrid	IFA	1-89	General population	63/1825(3.45%)	Sex (male)	Lledó et al 2004[45]
	Navarra	ELISA	15-59	General population	63/1429(4.41%)	occupational categories (livestock farmers), the type of cattle (north zone)	Oteiza et al 2011[46]
	Guadalajara Province	IFA	Adults	Forestry workers	7/100(7.00%)	Sex (male), age (older), contact with animals (rodents and foxes), arthropod bites	Lledó et al 2019[47]
	Nationwide	ELISA	N/A	Potential *Bb* infection population	432/2842 (15.20%)	live in rural, mountain area	Vázquez-López et al 2018[48]

Country	Region	Method	Age(y)	Group	Positive/Total (%)	Determined risk factors	Author year
Northern Europe							
Finland	Helsinki	ELISA, conf. WB	3-82	Patients with idiopathic facial paresis	6/42(14.29%)	children with FP	Jäämaa et al 2003[49]
				Patients with other neurological disorders	1/42(2.38%)		
	Nationwide	ELISA	15-86	All population	199/994(20.02%)	Age (older), unemployment, self-reported perception of feeling (unhealthy), heart valvular disease, areas (south, central, east)	Cuellar et al 2020[50]
	Nationwide	ELISA	29-97	All population	86/2000(4.30%)	Sex (male), age (older), regions (central, southern et al)	van et al 2017[51]
Norway	Agder County	ELISA	N/A	Patients with PN symptoms	43/209(20.57%)	N/A	Mygland et al 2006[52]
				Healthy blood donors	45/247(18.22%)		
	Agder County	ELISA	18-69	All population	680/2968(22.91%)	Sex (male), age (older), less frequently moderate to severe somatic symptom load	Thortveit et al 2020[53]
	Sogn, Fjordane county	ELISA, conf. WB	19-69	Healthy blood donors	147/1213(12.12%)	Sex (male), age (older), number of tick bites, cat or dog owners (no)	Hjetland et al 2014[54]
Sweden	South-eastern	ELISA, conf. WB	≥18	Tick-bitten population	11/341(3.23%)	N/A	Fryland et al 2011[55]
	South-eastern	ELISA	18–70	Blood donors	133/573(23.21%)	Sex (male), age (60–69 years)	Johansson et al 2017[56]
Denmark	Southern Jutland	CLIA	19-67	Blood donors	27/344(7.84%)	tick bite, cat ownership(reverse correlation)	Jensen et al 2021[57]
Eastern Europe							
Romania	13 districts	PHA, conf. WB	10-79	Healthy blood donors	69/1598(4.32%)	Profession (exposed to tick bites)	Hristea et al 2001[58]
				Forestry workers	99/1068(9.27%)	Age (older than 55 years), history of tick bite	
	Alba, Bistrița-Năsăud, Cluj, Maramureș, Sălaj, Satu-Mare	ELISA, conf. WB	18-65	Blood donors	49/1200(4.08%)	Age (older than 56 years), professional activities outdoors	Kalmár et al 2021[59]
Ukraine	Ternopil	ELISA, conf. WB	1-99	Potential *Bb* infection population	43/124(34.68%)	N/A	Shkilna et al 2021[60]
				Patients with PTB	5/79(6.33%)		
Central Europe							
Slovakia	Nationwide	IFA/ELISA, conf.	N/A	Blood donors	32/250(12.80%)	N/A	Bazovska et al 2005[61]
	Eastern	ELISA, conf. WB	12-93	Potential *Bb* infection population	67/537(12.48%)	N/A	Bušová et al 2018[62]
	Košice	ELISA	Adults	Patients with possible symptoms of LB	31/126(24.60%)	Sex (female), age (older)	Bušová et al 2018[63]
				Gardeners and soldiers working in afforested areas	29/135(21.48%)	Sex (female)	

Country	Region	Method	Age(y)	Group	Positive/Total (%)	Determined risk factors	Author year
Czechia	Prague	ELISA	1-99	Psychiatric Patients	332/926(35.85%)	N/A	Hájek et al 2002[64]
				Consecutive healthy subjects	163/884(18.44%)		
	Prague	ELISA, conf. WB	Adults	Subjects with normal left ventricular systolic function	7/50(14.00%)	N/A	Kuchynka et al 2016[65]
Austria	East and north Tyrol	ELISA, conf. WB	18-61	Healthy blood donors	103/1010(10.20%)	Previous tick bite, sex (male), age (older), outdoor	Sonnleitner et al 2015[18]
Poland	South-eastern	ELISA	N/A	Forestry workers	46/113(40.71%)	forestry workers, age (older)	Cisak et al 2005[66]
				Urban blood donors	4/56(7.14%)		
	Eastern and central	ELISA	18-91	Farmers	964/3597(26.80%)	Age (older), living in the country for more than 10 years, living close to forest, spending time in the forest or its vicinity for more than 6 h daily, experienced tick bite, positive result of test for borreliosis, diagnosed or suspected borreliosis	Zając et al 2017[67]
	Lublin Province	ELISA, conf. WB	17-80	Individuals exposed to vector ticks	57/150(38.00%)	N/A	Pańczuk et al 2019[68]
	Southern Podlasie	ELISA, conf. WB	16-20	Adolescents bitten by tick	5/128(3.91%)	duration of tick	Pańczuk et al 2014[69]
	Lower Silesia	ELISA, conf. WB	21-67	Forest workers	142/646(21.98%)	Age (older), forest type (deciduous or mixed-deciduous forests)	Kiewra et al 2018[70]
	Western	ELISA, conf. WB	29-65	Forest rangers	18/48(37.50%)	N/A	Bura et al 2018[71]
	West Pomeranian region	ELISA, conf. IB	N/A	LB-like symptoms population	65/80(81.25%)	N/A	Wojciechowska et al 2021[72]
				General/Healthy population	2/22(9.09%)		
Western Asia							
Jordan	Nationwide	ELISA	6-91	Genera population	96/824(11.65%)	Sex (female), age (<20 or >60 years), living in south-eastern area	Obaidat et al 2020[73]
Turkey	Izmir	ELISA, conf. WB	20-74	Patients with Behçet's disease	4/30(13.33%)	N/A	Onen et al 2003[74]
				Patients with rheumatoid arthritis	7/31(22.58%)	N/A	
				Healthy subjects	4/31(12.90%)	N/A	
	Düzce	IFA/ELISA, conf.WB	14-92	Forestry workers and farmers	4/349(1.15%)	forestry workers and farmers, age (10-20 years), sex (male), animal contact	Kaya et al 2008[75]
				Healthy blood donors	0/193(0.00%)		
	Samsun	ELISA, conf. WB	5-81	All population	14/419(3.34%)	Age (>45 years), sex(female), rural areas, altitude of ≥ 400 meters, locations suitable for wild boar and rabbits, dog owners	Aslan et al 2012[76]

Country	Region	Method	Age(y)	Group	Positive/Total (%)	Determined risk factors	Author year
	Trabzon	ELISA, conf. WB	20-79	All population	128/884(14.48%)	Age (≥70 years), educational level (low)	Cora et al 2017[77]
	Bolu	ELISA, conf. WB	N/A	All population	16/196(8.16%)	N/A	Bucak et al 2016[78]
	Düzce	ELISA, conf. WB	18-70	Forest villages	15/193(7.77%)	Materials used in house construction (mud-brick houses)	Akar et al 2019[79]
	Erzincan	ELISA, conf. WB	N/A	General population	8/368(2.17%)	Living area (rural)	Cikman et al 2018[80]
Eastern Asia							
China	Jilin, Qinghai, Gansu, Shanxi, Hunan, Guizhou, Beijing, Henan Provinces	IFA	5-75	People Living in forest areas	379/ 3669(10.33%)	N/A	Hao et al 2013[81]
	North-eastern	IFA	1-80	Tick-bitten population	33/180(18.33%)	N/A	Liu et al 2019[82]
	Anhui province	Protein Biochip	1-84	Patients with neuroborreliosis	52/56(92.86%)	N/A	Huang et al 2017[83]
	Hainan Province	IFA, conf. WB	5-92	Patients with suspected LB	28/252(11.11%)	Sex (male), age (younger than 20, older than 81years)	Wen et al 2021[84]
Korea	South	IFA	Adults	Forestry workers	53/655(8.09%)	Eating meals in woodland, raising dogs outside homes	Acharya et al 2021[85]
Japan	Saitama prefecture	ELISA	16-74	Forestry workers	25/80(31.25%)	Tick bites	Ikushima et al 1999[86]
Southern Asia							
India	South	ELISA, conf. WB	18-76	Forestry workers	14/472(2.97%)	exposure to tick bites, occupation (field workers)	Babu et al 2020[87]
South-eastern Asia							
Malaysia	8 states in Peninsular Malaysia	ELISA	5-83	Indigenous people	73/904(8.08%)	Village of residence (Tumboh Hangat/ Lubuk Legong/ Semanggar/ Dusun Kubur)	Khor et al 2019[88]
Oceania							
Australia	Sydney, Brisbane, Hobart, Devonport, Burnie	ELISA, conf. WB	N/A	All population	11/152 (7.24%)	N/A	Best et al 2019[2]
				Blood donor	0/308(0.00%)		
Africa							
Kenya	Nationwide	ELISA	1-99	Patients with acute febrile illness	375/1033(36.30%)	Age (younger)	Nyataya et al 2020[89]

Abbreviations: ELISA, Enzyme-Linked Immunosorbent Assay, conf. WB, confirmatory Western Blot, N/A, not available, IFA, indirect fluorescent antibody test, CLIA, Chemiluminescence Immunoassay PHA, Passive Hemagglutination Assays, LB, Lyme borreliosis, CES-D, Center for Epidemiologic Studies Depression scale, LTDs,

long-term-diseases, PN, chronic polyneuropathy, ALS, amyotrophic lateral sclerosis, PTB, pulmonary tuberculosis, Allocation of countries to geographical regions corresponds to the WHO region.

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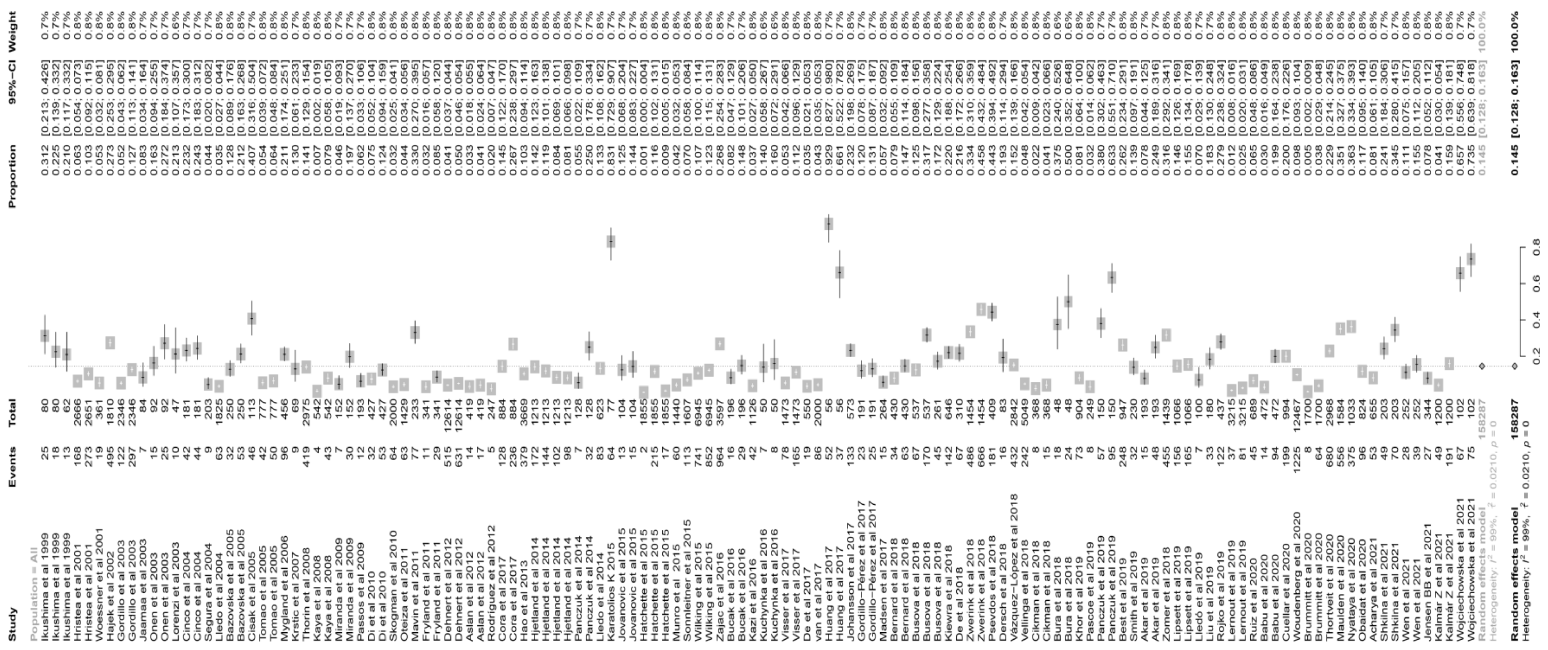
Supplementary appendix 6. Details of the risk of bias assessment with overall study-level risk of bias.

Source	Selection			Comparability	Outcome			Total score (%)	Overall quality
	Representativeness of the sample	Sample size	Ascertainment of exposure	Confounders	Ascertainment of outcome	Non respondents	Statistical analysis		
Cinco et al 2004		*	*	*	*	*	*	60	Moderate
Segura et al 2004	*	*	*	*			*	50	Moderate
Ruiz et al 2020	*	*	*	*	*	*	*	70	High
Lledo et al 2004	*	*	*	*	*	*	*	70	High
Gordillo et al 2003	*	*	*	*	*	*	*	70	High
Jaamaa et al 2003	*	*	*	**	*	*	*	80	High
Onen et al 2003	*	*	*	*	*	*	*	70	High
Lorenzi et al 2003		*	*	*	*	*	*	60	Moderate
Hristea et al 2001	*	*	*	*	*		*	50	Moderate
Hajek et al 2002		*	*	**	*	*	*	70	High
Woessner et al 2001		*	*	*	**	*	*	70	High
Ikushima et al 1999	*	*	*	*	*	*	*	70	High
Miranda et al 2009	*	*	*	*	*	*	*	70	High
Passos et al 2009		*	*	*	*	*	*	60	Moderate
Thorin et al 2008	*	*	*	*	**	*	*	80	High
Kaya et al 2008	*	*	*	**	*	*	*	80	High
Krstic et al 2007		*	*	*		*	*	50	Moderate
Mygland et al 2006	*	*	*	*	*	*	*	70	High
Bazovska et al 2005		*	*	*		*	*	50	Moderate
Cisak et al 2005		*	*	**	*	*	*	70	High
Tomao et al 2005	*	*	*	**	**	*	*	90	High
Karatolios K 2015	*	*	*	*	*	*	*	70	High
Hjetland et al 2014	*	*	*	*	*	*	*	70	High
Hao et al 2013	*	*	*	*	*	*	*	70	High
Dehnert et al 2012		*	*	*	*	*	*	60	Moderate
Aslan et al 2012	*	*	*	*	*	*	*	70	High
Rodríguez et al 2012	*	*	*	*	*	*	*	70	High

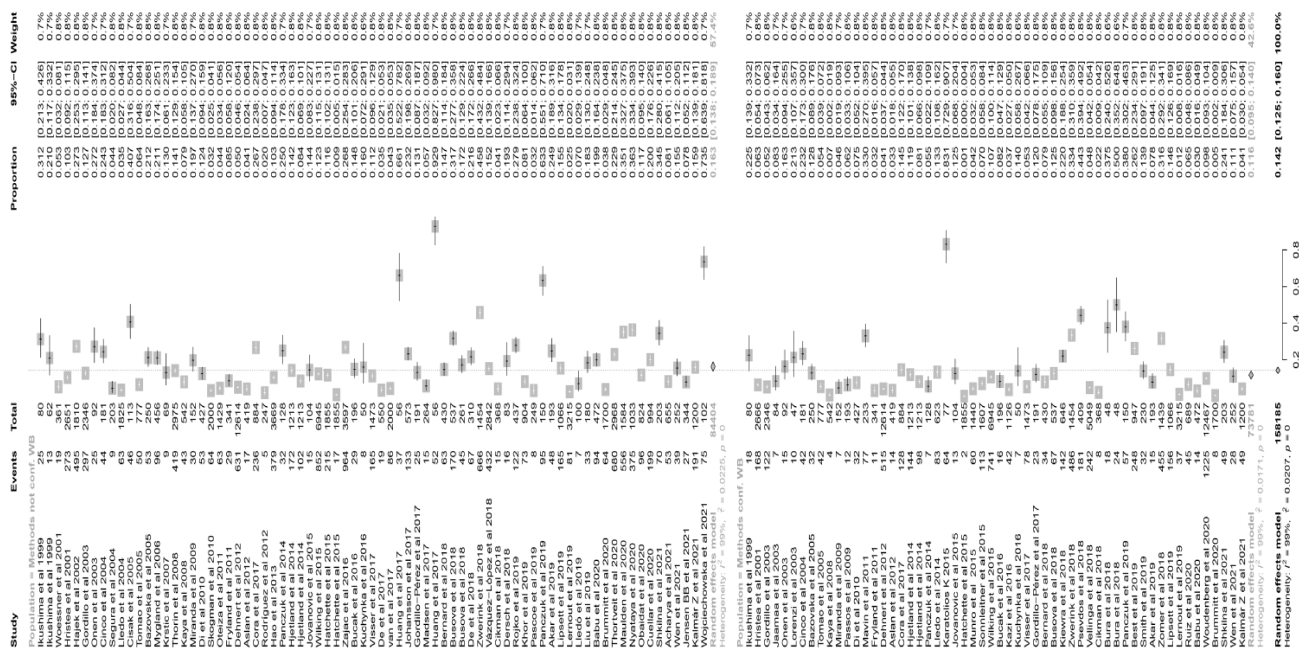
Oteiza et al 2011	*	*	*	*	*	*	*	70	High
Di et al 2010		*	*	**	*	*	*	70	High
Skogman et al 2010	*	*	*	**	*	*	*	80	High
Bernard et al 2018		*	*	*	**	*	*	70	High
Cora et al 2017	*	*	*	*	*	*	*	70	High
Visser et al 2017		*	*	**	*	*	*	70	High
De et al 2017		*	*	*	*	*	*	60	Moderate
Zajac et al 2016	*	*	*	*	*	*	*	70	High
Bucak et al 2016	*	*	*	*	*	*	*	70	High
Kazi et al 2016		*	*	*	**	*		60	Moderate
Kuchynka et al 2016		*	*	*	*	*	*	60	Moderate
Jovanovic et al 2015		*	*	**	**	*	*	80	High
Hatchette et al 2015		*	*	*	*	*	*	60	Moderate
Munro et al 2015	*	*	*	*	*	*	*	70	High
Sonnleitner et al 2015	*	*	*	*	*	*	*	70	High
Wilking et al 2015		*	*	*	*	*	*	60	Moderate
Panczuk et al 2014		*	*	*	**	*	*	70	High
Khor et al 2019		*	*	*	*		*	50	Moderate
Babu et al 2020	*	*	*	*	**		*	70	High
Lledo et al 2014		*	*	*	*	*	*	60	Moderate
Pascoe et al 2019	*	*	*	*	*	*	*	70	High
Panczuk et al 2019		*	*	*	*	*	*	60	Moderate
Best et al 2019		*	*	*		*	*	50	Moderate
Smith et al 2019	*	*	*	*	*	*	*	70	High
Akar et al 2019	*	*	*	*	*	*	*	70	High
Busova et al 2018		*	*	*	*	*	*	60	Moderate
Busova et al 2018		*	*	*	*	*	*	60	Moderate
Zomer et al 2018		*	*	*	**	*	*	70	High
Kiewra et al 2018		*	*	*	**	*	*	70	High
De et al 2018	*	*	*	*	*	*	*	70	High
Mavin et al 2011		*	*	*	**	*	*	70	High
Cuellar et al 2020	*	*	*	*	*	*	*	70	High
Lipsett et al 2019	*	*	*	*	**	*	*	80	High
Woudenberg et al 2020	*	*	*	*	**	*	*	80	High
Fryland et al 2011	*	*	*	*	**	*	*	80	High
Acharya et al 2021	*	*	*	*	*	*	*	70	High

Lledó et al 2019	*	*	*	*	*	*	*	70	High
Liu et al 2019	*	*	*	*	*	*	*	70	High
Zwerink et al 2018		*	*	*	**	*	*	70	High
Rojko et al 2019		*	*	*	*	*	*	60	Moderate
van et al 2017	*	*	*	*	**	*	*	80	High
Psevdos et al 2018		*	*	*	*	*	*	60	Moderate
Dersch et al 2018	*	*	*	**	**	*	*	90	High
Brummitt et al 2020		*	*	*	**	*	*	70	High
Thortveit et al 2020	*	*	*	*	*	*	*	70	High
Huang et al 2017		*	*	*	*	*	*	60	Moderate
Lernout et al 2019	*	*	*	*	**	*	*	80	High
Johansson et al 2017	*	*	*	*	*	*	*	70	High
Maulden et al 2020	*	*	*	*	**	*	*	80	High
Nyataya et al 2020	*	*	*	*	*	*	*	70	High
Shkilna et al 2021		*	*	*	**	*	*	70	High
Vázquez-López et al 2018		*	*	*	*	*	*	60	Moderate
Vellinga et al 2018		*	*	*	*	*	*	60	Moderate
Cikman et al 2018	*	*	*	*	**	*	*	80	High
Gordillo-Pérez et al 2017	*	*	*	*	*	*	*	70	High
Wen et al 2021	*	*	*	*	*	*	*	70	High
Obaidat et al 2020	*	*	*	*	*	*	*	70	High
Bura et al 2018	*	*	*	*	*	*	*	70	High
Madsen et al 2017	*	*	*	*	*	*	*	70	High
Jensen et al 2021	*	*	*	*	*	*	*	70	High
Kalmár et al 2021	*	*	*	*	*	*	*	70	High
Wojciechowska et al 2021	*	*	*	*	**	*	*	80	High

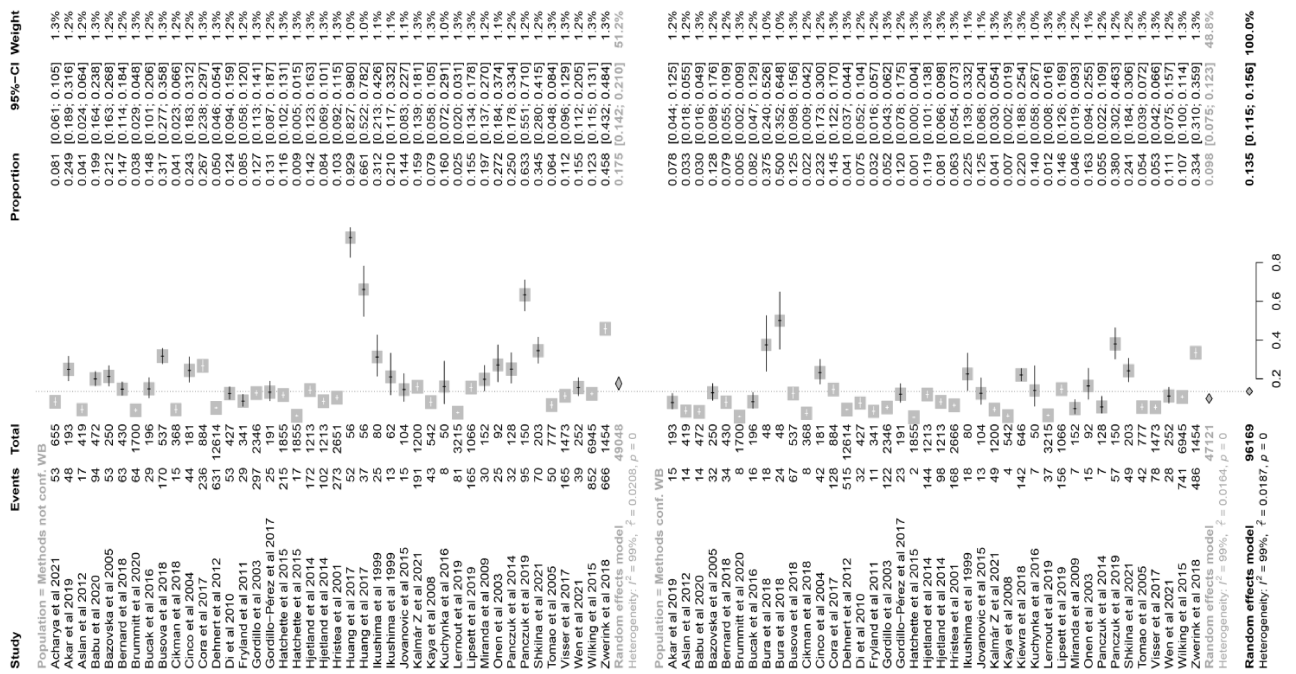
Supplementary appendix 7. Forest plot of antibodies to *Borrelia burgdorferi sensu lato* prevalence among included 89 studies.



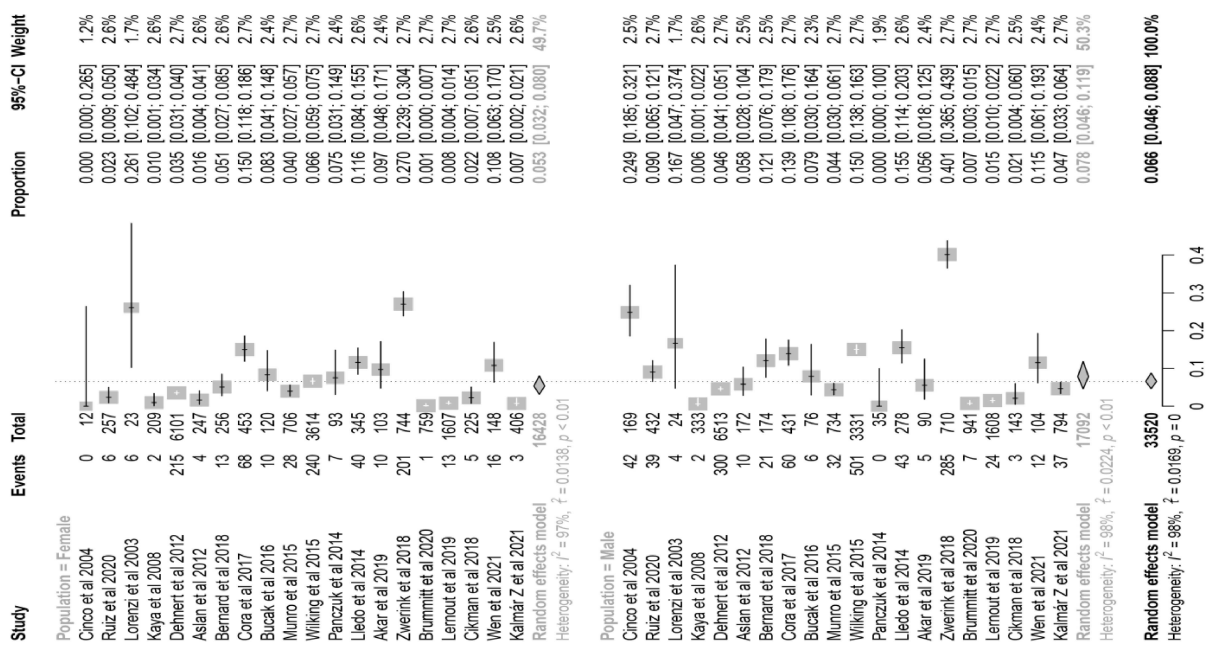
Supplementary appendix 8. Forest plot of antibodies to *Borrelia burgdorferi sensu lato* prevalence among people who examined by methods conf. WB relative to methods not conf. WB of included 89 studies.



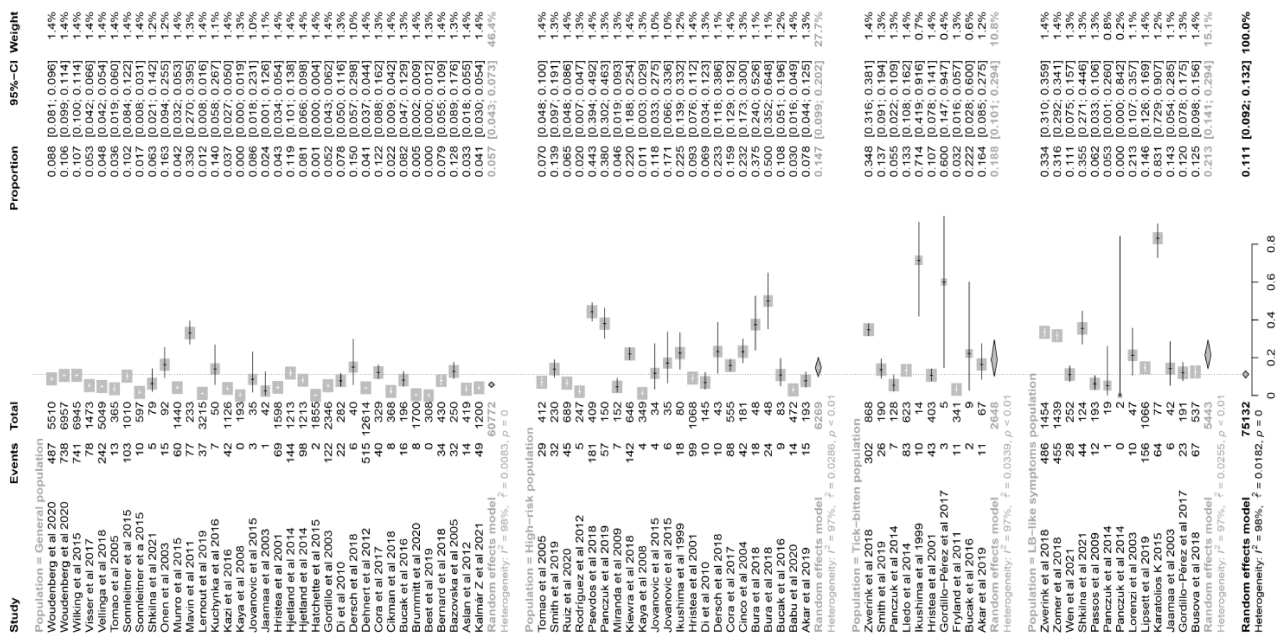
Supplementary appendix 9. Forest plot of antibodies to *Borrelia burgdorferi sensu lato* prevalence among 40 extracted studies that reported people who examined by methods conf. WB Relative to methods not conf. WB.



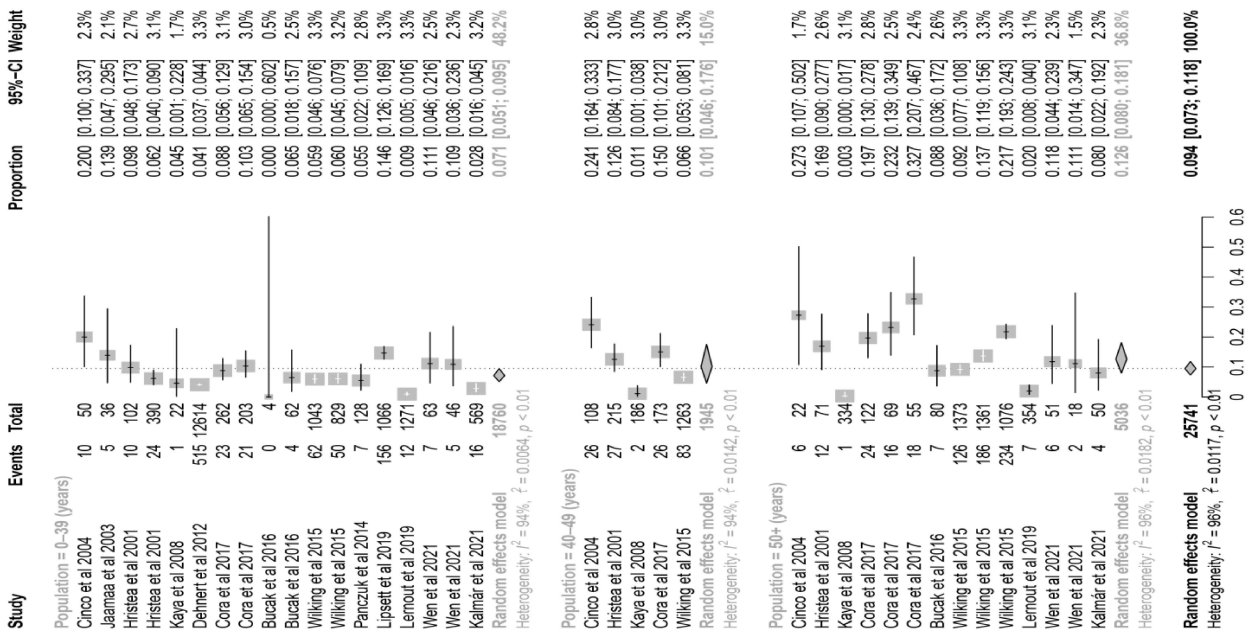
Supplementary appendix 10. Forest plot of antibodies to *Borrelia burgdorferi sensu lato* prevalence among female relative to male of included 58 studies.



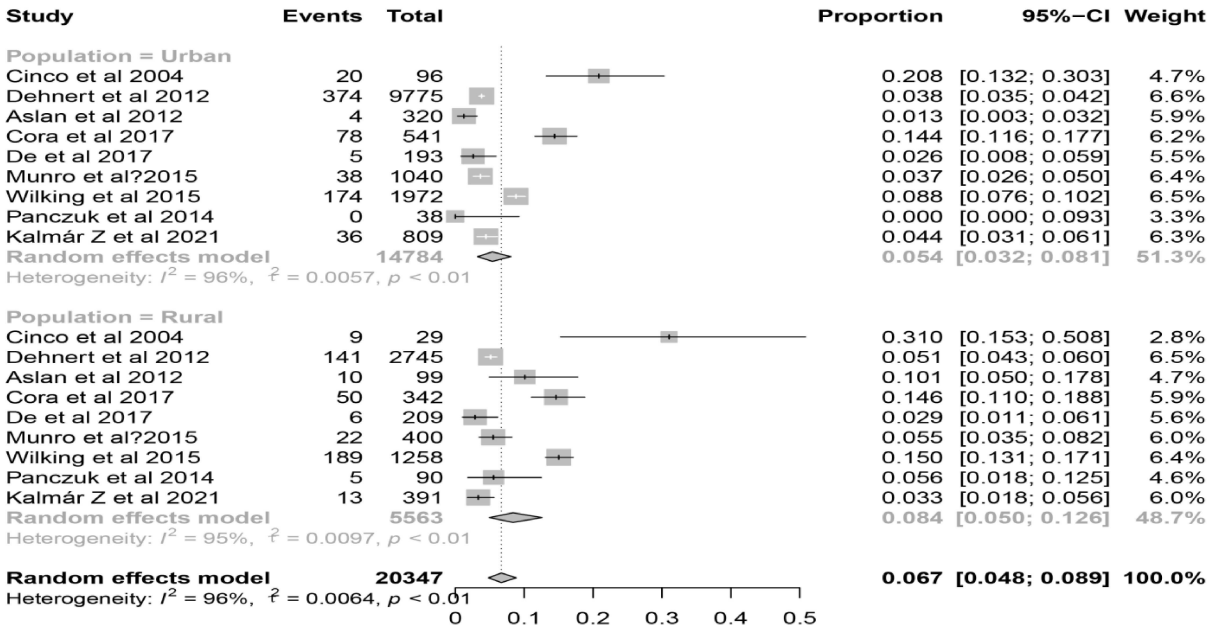
Supplementary appendix 11. Forest plot of antibodies to *Borrelia burgdorferi sensu lato* prevalence among different people of included 58 studies.



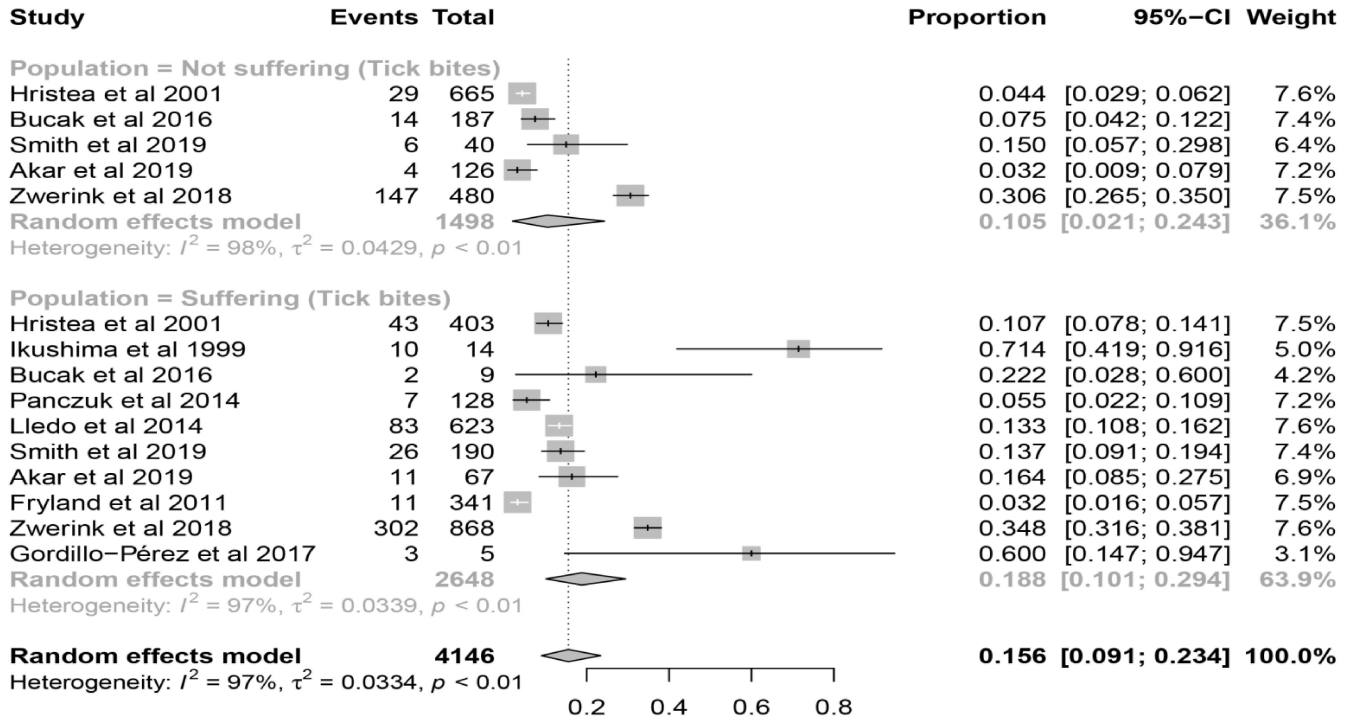
Supplementary appendix 12. Forest plot of antibodies to *Borrelia burgdorferi sensu lato* prevalence among different age groups of included 58 studies.



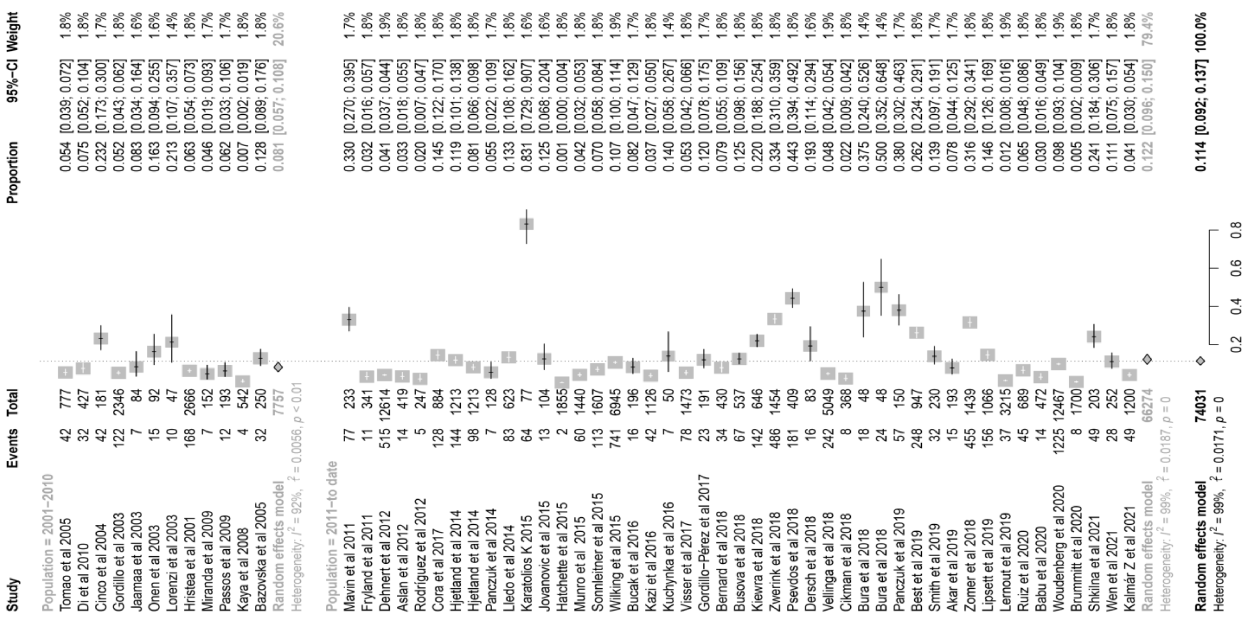
Supplementary appendix 13. Forest plot of antibodies to *Borrelia burgdorferi sensu lato* prevalence among residence in rural relative to residence in urban of included 58 studies.



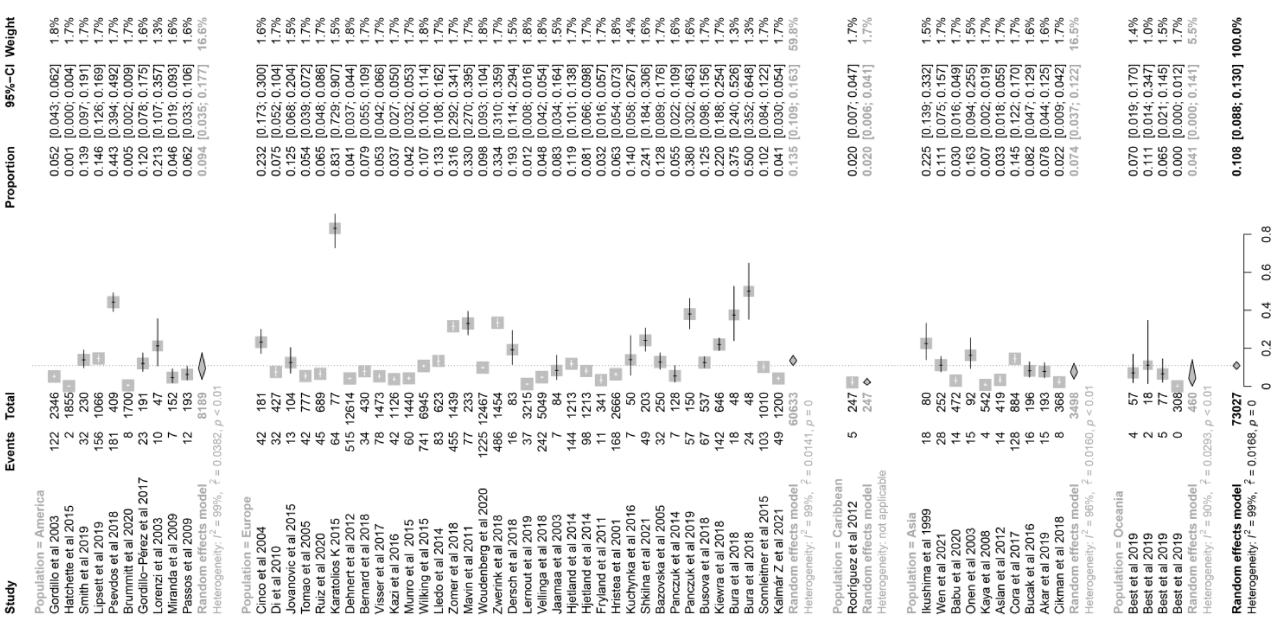
Supplementary appendix 14. Forest plot of antibodies to *Borrelia burgdorferi sensu lato* prevalence among suffering tick bites population relative to not suffering tick bites population of included 58 studies.



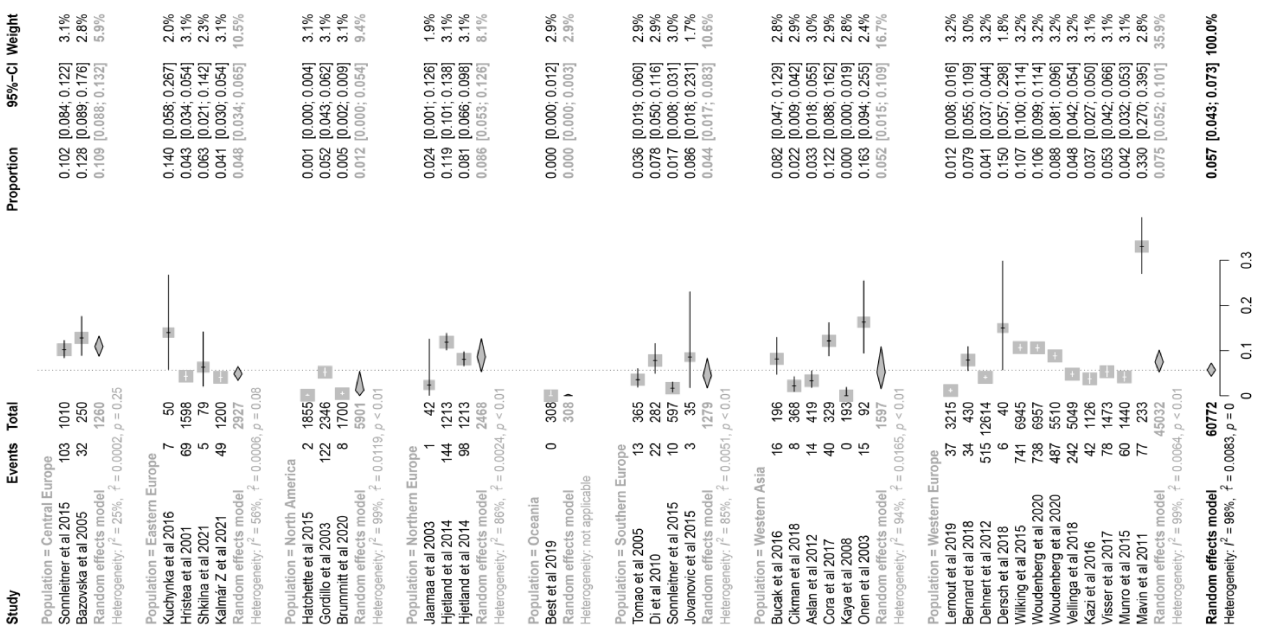
Supplementary appendix 15. Forest plot of antibodies to *Borrelia burgdorferi sensu lato* prevalence among people who examined in 2011-to date relative to people who examined in 2001-2010 of included 58 studies.



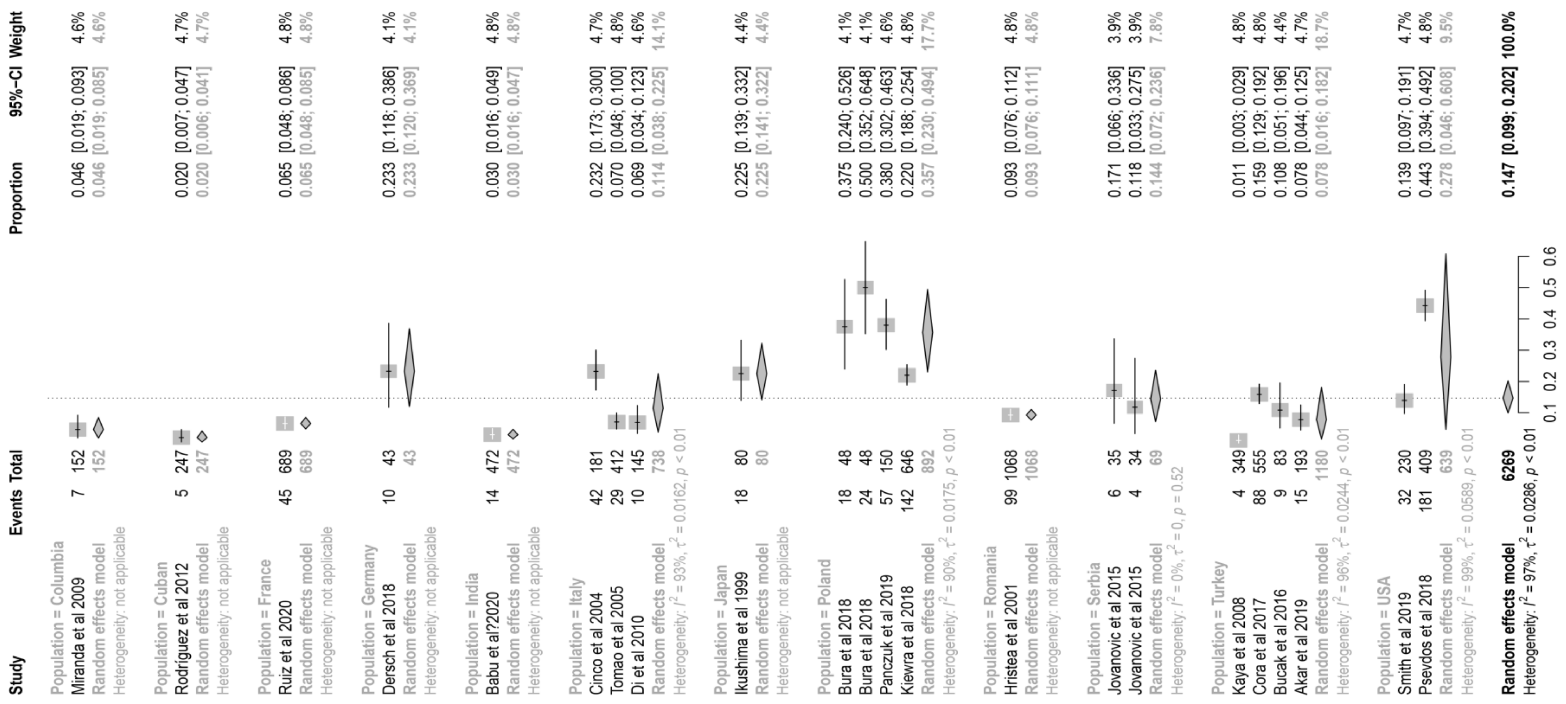
Supplementary appendix 16. Forest plot of antibodies to *Borrelia burgdorferi sensu lato* prevalence among different continent of included 58 studies.



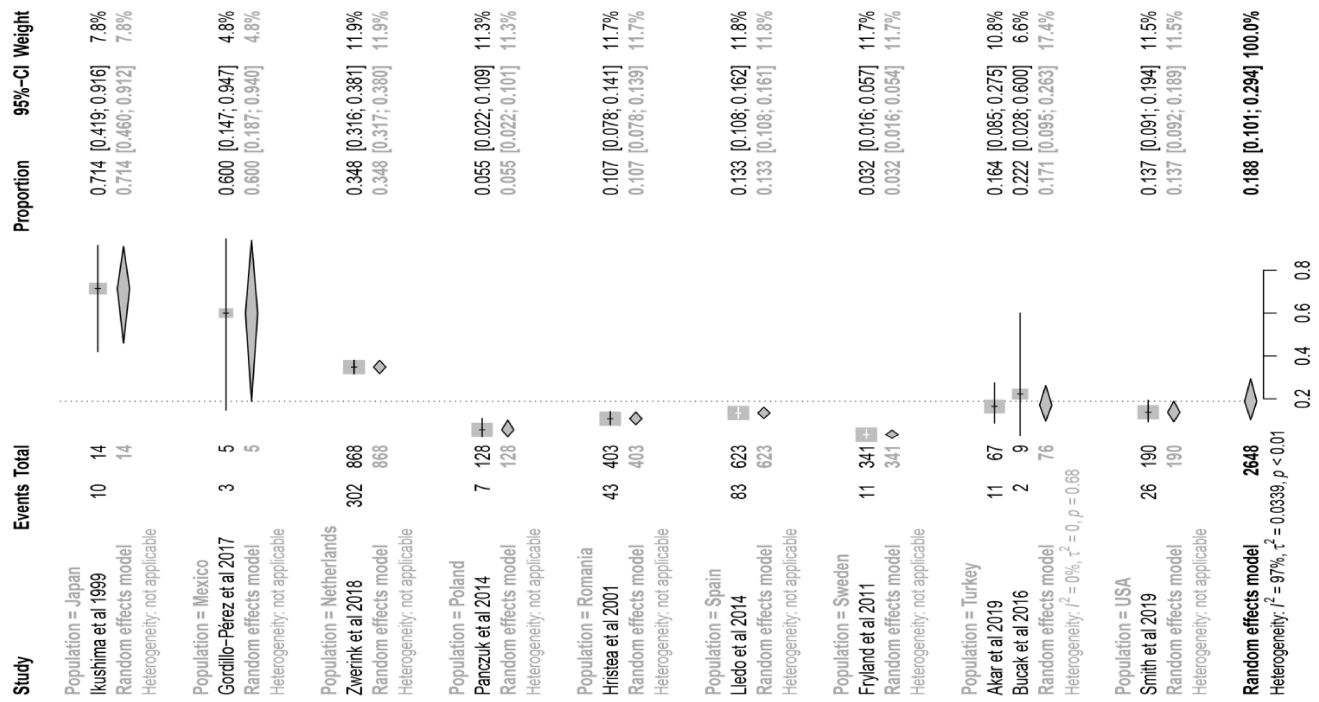
Supplementary appendix 17. Forest plot of antibodies to *Borrelia burgdorferi sensu lato* prevalence among different continent of General population in included 58 studies.



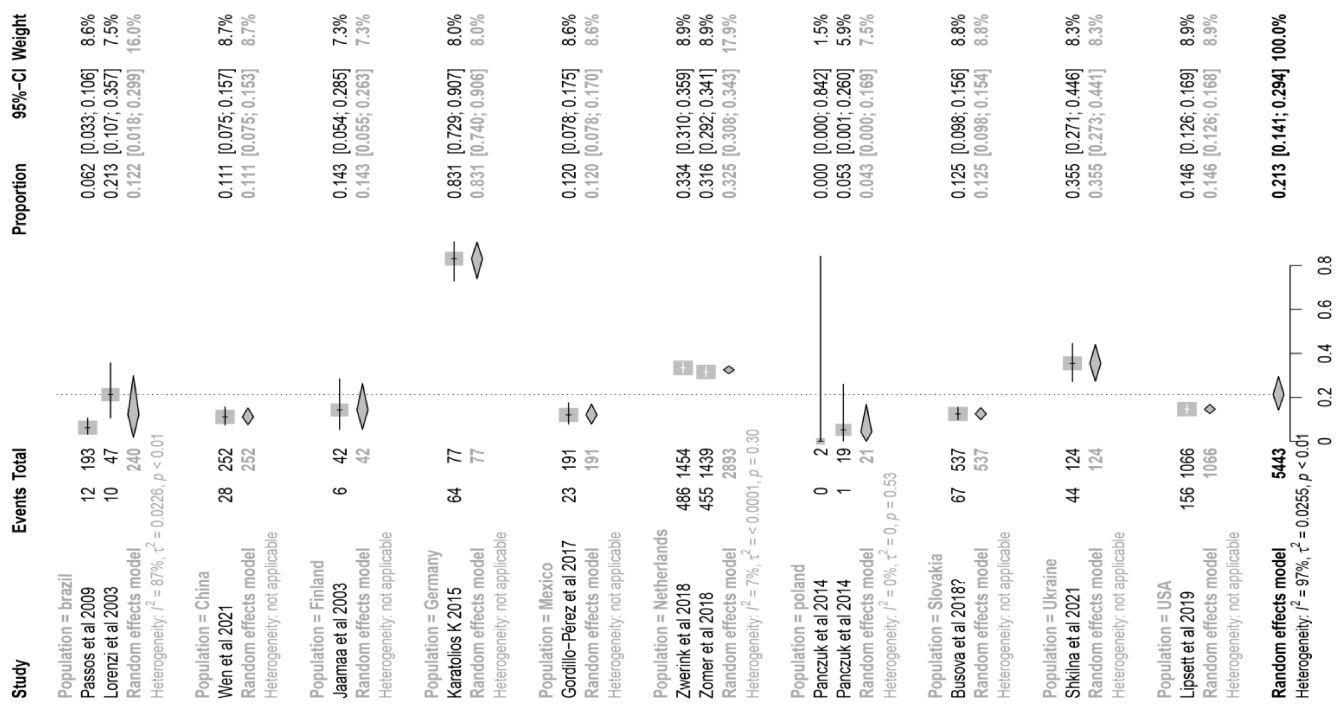
Supplementary appendix 18. Forest plot of antibodies to *Borrelia burgdorferi sensu lato* prevalence among different continent of High-risk population in included 58 studies.



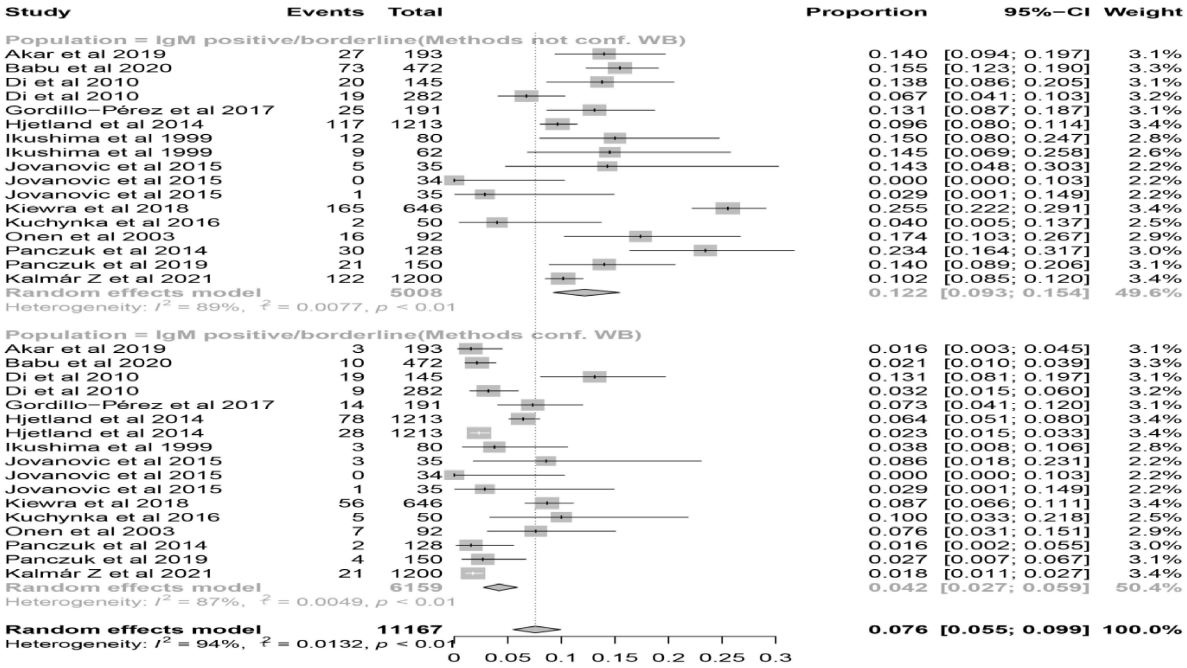
Supplementary appendix 19. Forest plot of antibodies to *Borrelia burgdorferi sensu lato* prevalence among different continent of Tick-bitten population in included 58 studies.



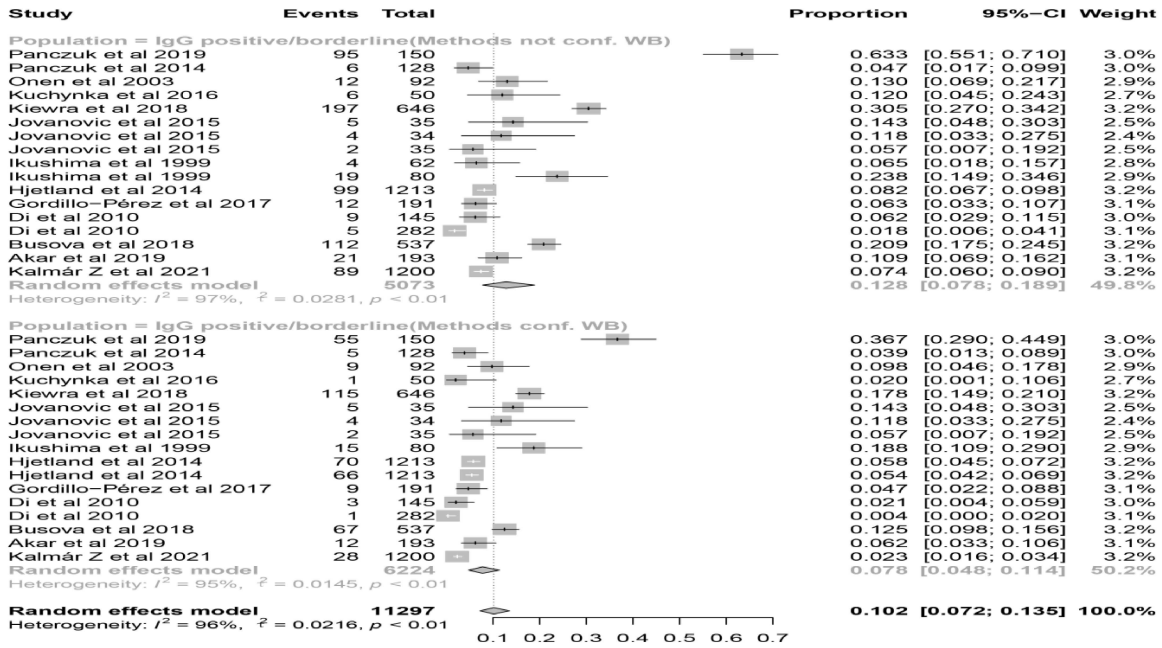
Supplementary appendix 20. Forest plot of antibodies to *Borrelia burgdorferi sensu lato* prevalence among different continent of LB-like symptoms population in included 58 studies.



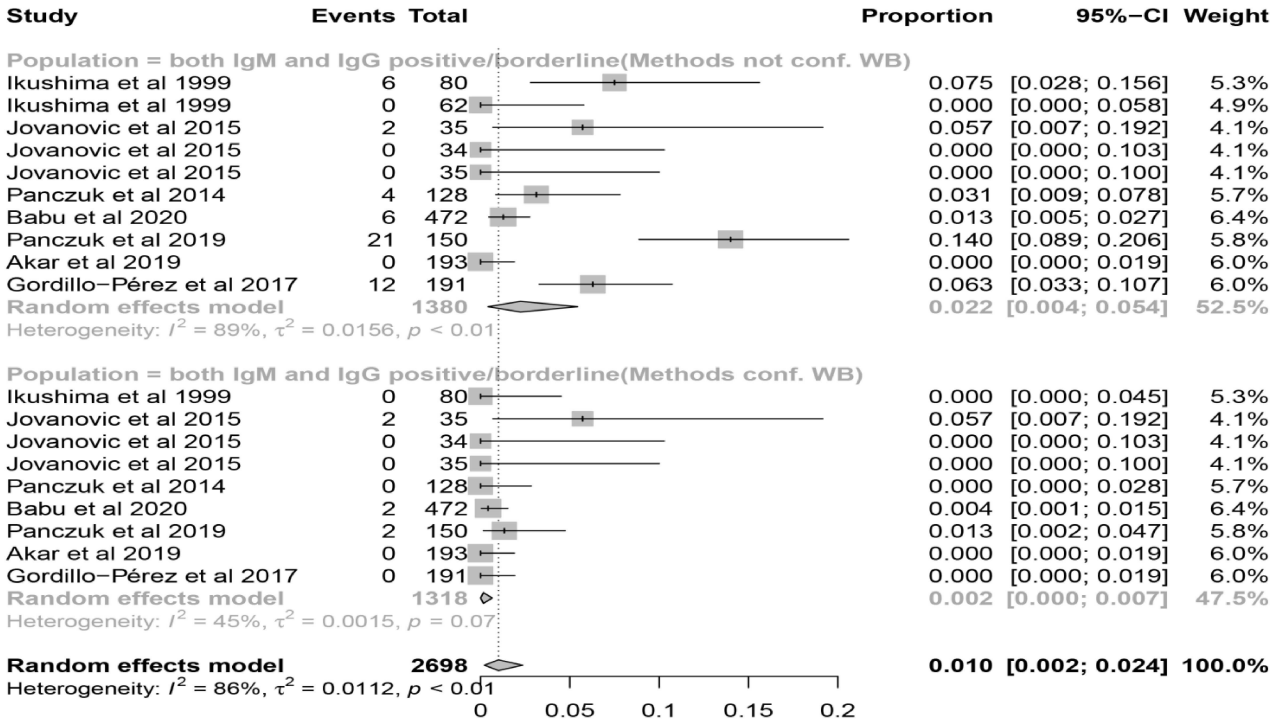
Supplementary appendix 21. Forest plot of specific anti-IgM antibody to *Borrelia burgdorferi sensu lato* prevalence among 40 extracted studies that reported people who examined by methods conf. WB Relative to methods not conf. WB.



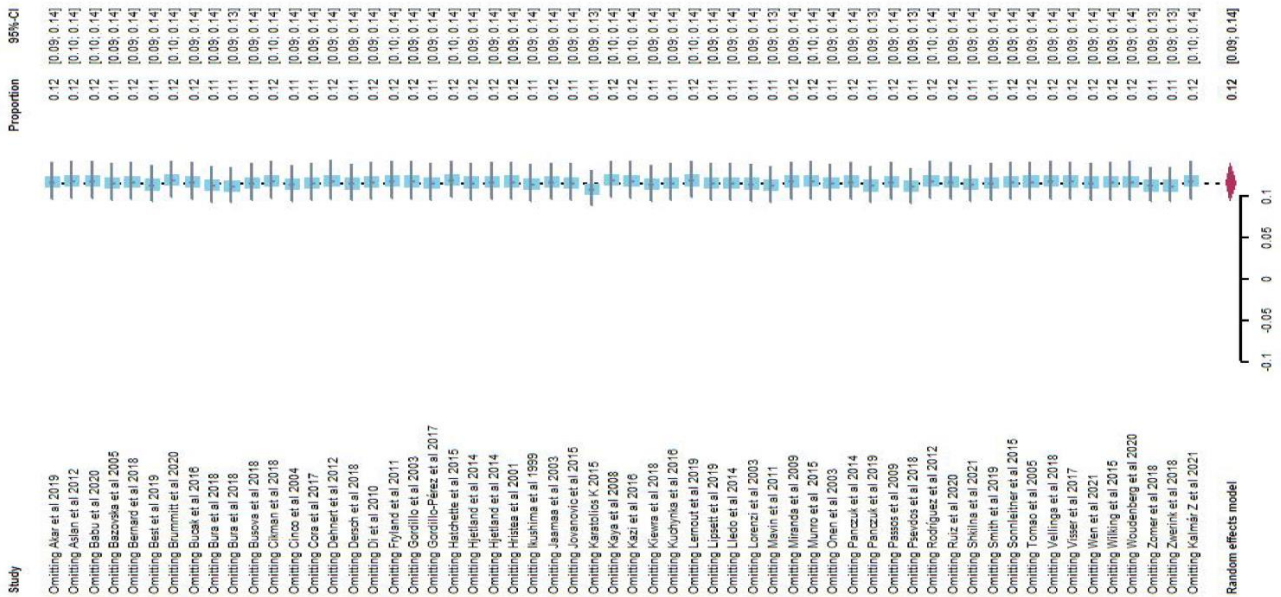
Supplementary appendix 22. Forest plot of specific anti-IgG antibody to *Borrelia burgdorferi sensu lato* prevalence among 40 extracted studies that reported people who examined by methods conf. WB relative to methods not conf. WB.



Supplementary appendix 23. Forest plot of both specific anti-IgM and anti-IgG antibodies to *Borrelia burgdorferi sensu lato* prevalence among 40 extracted studies that reported people who examined by methods conf. WB relative to methods not conf. WB.



Supplementary appendix 24. Sensitivity analysis of the overall *Borrelia burgdorferi sensu lato* seroprevalence.



Supplementary appendix 25. Forest plot of antibodies to *Borrelia burgdorferi sensu lato* prevalence among included 58 studies examined by methods conf. WB.